

Sentiment Analysis in Music

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Introduction

- Describe why your project is interesting. Describe the motivation from a personal learning perspective.
 - We decided to work with music because we are all very interested in the music industry. As such, we wanted to see how well we can classify different songs based on their genre using machine learning techniques. In addition, we wanted to see differences in classification based on using metadata vs. lyrics, as well as how unsupervised models would group unlabeled songs together.
- Project category (Music, Vision, Finance, Life Sciences, Natural Language...)
 - We will be focusing on music, specifically genre detection and sentiment analysis. Natural language processing will also be involved in the lyrical analysis section of our project.
- Describe what you are planning to do.

We plan to first query the Spotify API for our data, using it mainly for audio features, then the Genius API for corresponding lyrics. Rapidfuzz may be used to match inconsistencies between song titles from the two APIs. Lyrics will go through TF-IDF, Word2vec, or some other ways of NLP, before combination with audio features. Before model selection, we will map correlation heatmap of audio features, plot distributions of features by genre, and make word clouds or frequency analysis for lyrics.

We will use linear, ridge, or lasso regression as a baseline model for prediction of genre. We will use XGBoost and KNN to also compare against those base models. Unsupervised techniques, including K-means clustering, could be used to first pit against aforementioned models and also to produce new features that can then be used again to increase the accuracy of our baseline models.

We will then split the data, train, and evaluate the efficiency of our results using F1-scores and confusion matrices. There may be some hyperparameter tuning. We will derive insights from our data, including feature importance from XGBoost, and coefficients from regressions. We will make visualizations of this. We will also seek to answer questions like if fast songs are all energetic, or which features most separate genres.

- What would be the successful outcome of your project?

70–85% genre classification accuracy by any model would be our target. Ample visualization would be good eventually for presentation. It would also be good to answer questions like which feature contributed most to prediction.

Analysis plan

- Describe the ML model(s) to be used.

We plan to use linear regression / regularized regression as base models, KNN and XGBoost for our supervised learning techniques, and conduct unsupervised techniques as well to group similar songs. We also plan to use NLP models for lyrical data, such as LSTMs and the Transformer Architecture.

- Describe your data set, including the data source link. Describe its size and features (categorical and continuous).
 - o Genius Song Lyrics (Kaggle) - this dataset has about 5 million songs; features are categorical (title, artist, lyrics, language, tag (genre))
 - SOURCE:
<https://www.kaggle.com/datasets/carlosgdj/genius-song-lyrics-with-language-information>
 - o Spotify tracks dataset (Kaggle) - this dataset has about 114,000 tracks; features are continuous such as genre, popularity, danceability, energy, etc..
 - SOURCE:
<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>
 - o Spotify Songs (Kaggle) - this dataset has about 30,000 tracks; features are categorical: they are playlist_genre, playlist_sub_genre, songs
 - SOURCE:
<https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset>
 - o WASABI Song Corpus - this dataset has 2,000,000 songs; features are mixed categorical and continuous such as genre, mood, language, chords, tempo.
 - SOURCE: <https://github.com/micbuffa/WasabiDataset>

Contributors

- Describe what each team member will contribute to the project.
 - o Sam: Run unsupervised learning such as K-Means Clustering on the data to learn
 - o Cade: Scrape and preprocess all of the data for supervised and unsupervised tasks

- o Larry: Run all supervised models such as base models, XGBoost, etc.